

CPSTRIDE: A Threat Modeling Framework for Cyber-Physical Systems

Dallas Elleman¹[0009–0001–7490–2440], John Hale¹[0000–0002–5956–8093]

University of Tulsa, Tulsa OK 74104, USA
dallas-elleman@utulsa.edu, john-hale@utulsa.edu
<https://utulsa.edu/programs/cyber-security/>

Abstract. Cyber-physical systems requires threat models that account for cyber and physical vulnerabilities alike. We present the CPSTRIDE framework, which extends the classic STRIDE model with a novel Cyber-Physical Flow Diagram and updated Security Property and Threat definitions. We demonstrate CPSTRIDE’s utility by modeling threats beyond STRIDE’s capabilities, and employ LLM assistance to identify threats in an additive manufacturing context.

Keywords: Cyber-Physical Systems · Additive Manufacturing · Threat Modeling · STRIDE · Critical Infrastructure · LLM-assisted

1 Introduction

Cyber-physical systems (CPS) combine sensors, actuators, and computation to monitor and control physical processes and enable automation and efficiency. They are integral to distributed, adaptive, and data-driven decision-making and operations in nearly every critical sector, including transportation, health care, energy, and Industry 4.0 [12]. Additive manufacturing (AM) technologies are a quintessential example of CPS that merge digital and physical processes to overcome legacy manufacturing limitations [3]. Automation and connectivity confer many advantages in CPS, but also expose new attack paths that compromise safety-critical systems. In light of widespread and potentially catastrophic risks to infrastructure, services, and human safety, CPS threat modeling is imperative for their secure and resilient design, deployment, operation, and maintenance [5].

Threat modeling frameworks (e.g., STRIDE, MITRE ATT&CK, Cyber Kill Chain) are used widely in the design, modeling, and analysis of information and communications technologies such as computers, software applications, and the Internet. However, these frameworks lack the native capacity to model physical vulnerabilities in CPS, such as in plant and machinery, material flows, energetic processes, physical access controls and trust boundaries, and many other elements and interactions [10]. Our research fills this gap with a purpose-built framework for CPS threat modeling, comprising several primary contributions:

1. We present CPSTRIDE, a threat modeling framework for CPS that extends the STRIDE framework with updated security properties and threats.

2. We describe the Cyber-Physical Flow Diagram (CPFD), which transcends the STRIDE Data Flow Diagram (DFD) paradigm by modeling physical and cyber-physical elements and threats and distinguishing between processes and devices, and between flows and paths.
3. We demonstrate CPSTRIDE’s utility for CPS threat modeling - and its advantages over STRIDE - via LLM-assisted comparative threat modeling and identification on a representative AM system.

The remainder of the paper is organized as follows: Section 2 presents background information and previous work related to CPS and AM threats and STRIDE-based threat modeling. Section 3 details the CPSTRIDE framework. Section 4 describes our comparative threat modeling of an AM system using both CPSTRIDE and STRIDE. Section 5 compares and discusses the results. Section 6 suggests future directions and offers concluding remarks.

2 Background / Previous Work

2.1 Cyber-Physical System Threat Modeling with STRIDE

A substantial body of research is directed at re-purposing existing threat modeling frameworks for use in CPS; one popular choice is Microsoft’s STRIDE framework [13]. STRIDE’s iterative four-step process entails creating a Data Flow Diagram (DFD) for the system under study, identifying threats to each DFD element, investigating system vulnerabilities to threats, and prioritizing mitigations for vulnerabilities. DFDs use a set of five symbols that represent different types of software-relevant objects [8]. Yampolskiy et al. [17] extend STRIDE for application to uncrewed aerial systems (UAS). They note STRIDE’s insufficiency for differentiating CPS-relevant interactions, e.g., cyber vs. physical communications, and formulate an extended DFD or “xDFD” that enables modeling of “physical components, communication media, optional data flows, and physical signal” using four additional DFD elements. They address several threat modeling challenges in CPS, but do not expand STRIDE security property and threat definitions. Khan et al. [10] apply STRIDE to analyze a synchronous island microgrid electrical system, demonstrating the method’s expertise-informed brainstorming to identify cyber-attacks. They discuss the DFD’s shortcomings for representing physical and cyber-physical elements and processes in electric power systems, and opt to exclude components not susceptible to cyber attacks from their model. To our knowledge, there is no widely-recognized standard for representing physical system elements and threats using the STRIDE framework.

2.2 Additive Manufacturing Threat Modeling

AM systems combine digital design models and material feed stocks to produce physical objects, and as such are prime examples of CPS. Configuration and implementation details vary, but all AM systems exhibit the characteristic process chain illustrated by the entities and flows in Figure 1.

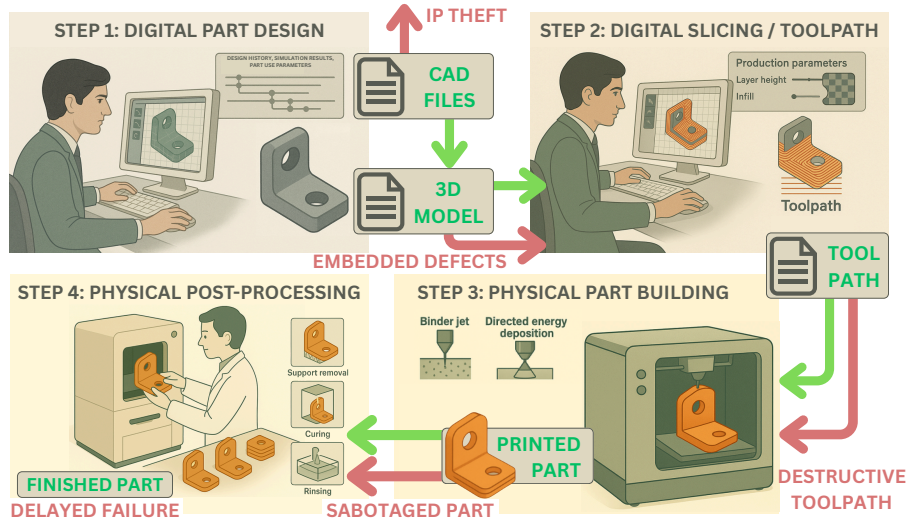


Fig. 1. The AM process chain, with benign and malicious cyber-physical flows

Using computer aided design (CAD) software, AM parts are digitally modeled and optimized through iterative simulation and analysis; the resulting CAD files include the part’s 3-D model, design history, and simulated performance information (if applicable). The 3-D model is then *sliced* into layers and a *toolpath* is generated that contains a set of machine instructions for physically building each part layer. This toolpath is then given to an AM machine, which executes each line of instructions to mechanically deposit and/or join feedstock one layer or small amount at a time. Once the toolpath file has been executed completely, the physical part may finally undergo post-processing, quality control testing, warehousing, shipping, and other product life cycle phases.

The AM process chain enables digital designs to be rapidly transformed into physical objects, with the unintended side effect of also allowing cyber-attacks to be rapidly translated into physical harms. AM systems are susceptible to regular cyber-security flaws as well as to unique attack vectors [4,6,7,11,18], some of which are illustrated in Figure 1. Sturm et al. identify four main opportunities for cyber-attacks to cross into the physical world: the CAD file, 3D model, toolpath file, and the physical machine [15]. However, numerous other cyber-attack vectors permit physical attack manifestations in AM machinery.

CAD files are the most information-dense digital artifacts produced by the AM process chain, containing a part’s geometric data, design history, simulated performance data, use parameters, and more, making them attractive targets for intellectual property (IP) theft and ransomware. Sabotage at this early stage may propagate data integrity and availability failures through the entire process chain [15]. 3-D model tampering can produce physical objects with hidden defects that are difficult or impossible to detect, causing unpredictable and dan-

gerous failure during use, as found by Belikovetsky [4]. Toolpath files are vectors for sabotaging manufactured objects, allowing theft of IP in the form of operational and manufacturing parameters, and can be modified to result in incorrect machine movement, dimension scaling, heating, feedrates, and other actions resulting in production defects and/or catastrophic equipment damage [18].

Supply chain attacks, as Yampolskiy et al. report, are difficult to defend against due to the sprawling ecosystem of third-party providers in AM systems [18]. Malware embedded in AM software, hardware, and firmware components could grant adversaries access to IP and even real-time operational control. Other attacks targeting mechanical AM system components actuators, and feedstocks can result in material composition, size, or form factor inconsistency. Gupta et al. examine supply chain vulnerabilities where material composition, purity, and microstructure might be attacked [7]. Side channels for AM system attack include acoustic, magnetic, and thermal emanations that allow the recreation or replication of AM parts without direct access to files. This type of technical data theft can be accomplished via smartphones and inexpensive surveillance equipment [1,2,9]. In addition, 3-D model files can carry hidden malicious payloads via steganographic embedding [16].

The unique nature of the digital-model-to-physical-object AM process chain exemplifies the need for a comprehensive cyber-physical threat framework. Next, we describe our approach to creating such a framework - one that is both rich enough to capture fine system and process detail, and abstract enough to describe diverse types of CPS. .

3 CPSTRIDE Framework Specification

We choose the mature and widely-used STRIDE model as a foundation due to its balance between specificity and generality for modeling cyber threats. While too data-centric and narrow for full description of CPS, we expand its coverage by examining its structure from first principles, starting with its four-step workflow:

1. Create a DFD model of the system.
2. Identify STRIDE threats, either per-element or per-interaction.
3. Investigate vulnerabilities to identified threats.
4. Plan mitigations for identified vulnerabilities.

Our changes to the STRIDE framework involve only the first two steps of the process; Steps 3 and 4 are not considered in our CPSTRIDE framework specification here, nor are they applied during side-by-side threat modeling of the AM system with both frameworks in Section 4. We begin with Step 1 by defining an expanded Cyber-Physical Flow Diagram (CPFD). For Step 2, we specify the CPS Security Properties and Threats required for threat identification.

3.1 CPFD: Cyber-Physical Flow Diagram

Drawing from Khan and Yampolskiy [10,17], and considering Shostack's [14] guidance to keep models "...aggressively simple to prioritize easy learning and use over expressiveness," we borrow five visual notation rules from DFDs:

1. All lines are solid, except for trust boundaries, which are dashed or dotted.¹
2. CPFDS can use color for additional information, but must not require it.
3. All elements should have a label.
4. Context diagram(s) for complex systems are optional.
5. Must, must not, should, should not are used per IETF norms.

Each existing DFD element has a name, a definition, an abbreviation, and a graphical symbol. Our process for developing the CPFD is to consider whether the *core meaning* of each existing DFD element definition can be expanded to include important physical and cyber-physical analogs in a CPS context. If so, we rename, redefine, and/or re-abbreviate the existing DFD element to include this context, and we retain its graphical symbol. Otherwise, we create a new element in the CPFD and strive for aggressive simplicity, à la Shostack [14]. Our resulting CPFD specification is found in Figure 2.

For example, the DFD *Interactor* (*I*) element (sometimes referred to as External Entity – EE – as in [10]) represents a person, code or website that interacts with the software system but is outside its control. The core meaning, “beyond system scope,” can be expanded to include CPS analogs such as physical raw material sources, gas or electric mains, cyber-physical supply chain providers, autonomous service robots, etc. We redefine a CPFD Interactor as “an entity that exchanges data, energy, or material with the CPS but remains outside its design scope and/or control boundary,” specify its cyber, physical, and cyber-physical abbreviations, and keep the graphical symbol (see Figure 2). Similarly we redefine the DFD *Trust Boundary* as “a virtual and/or physical zone of privileged access” and give it CPS abbreviation versions.

The DFD *Data Store* (*DS*) element represents data at rest (e.g., files, databases, registry keys) in software systems. This can be expanded to include CPS analogs at rest such as physical and cyber-physical materials, objects, and keys at rest. We wish to differentiate between the entity being stored and the storage medium as well as represent hierarchical or nested cyber-physical stores (e.g. digital files on a physical hard drive). Thus, we simplify the CPFD element name to *Store* and define it as “data, energy, or material at rest, distinct from its storage medium or container, with nesting allowed.” Abbreviations and graphical symbol changes can be found in Figure 2. We add a new element to the CPFD for physical and cyber-physical storage media and containers: *Device* (*D*).

Similarly, the *Data Flow* (*DF*) element represents data in motion (e.g., communication patterns and function calls); this can be expanded to include CPS analogs such as material flows, but we agree with Yampolskiy et al. [17] that CPS flow-enabling paths and media such as gas pipes, transmission lines, and radio frequency (RF) spectrum deserve first-class representation. We simplify the CPFD element name to *Flow*, define it as “data, energy, or material in motion, distinct from its enabling path, channel or medium,” modify its abbreviations and keep the graphical symbol, and add a new CPFD element to represent Flow enablers: *Link*.

¹ We extend the use of dashed lines for the two new CPFD elements: Links and Devices; see Figure 2.

Cyber-physical Flow Diagram (CPFD)			
<i>Cyber</i> indicates information, data, or signal; <i>physical</i> indicates material, energy, or force; <i>cyber-physical</i> indicates the integration or combination of the two. Each CPFD element is categorized as either cyber, physical, or cyber-physical depending on whether the element can reasonably be considered vulnerable to cyber and/or physical threats.			
Element name, abbreviation, and description	Graphical Symbol	Examples	C: Cyber P: Physical CP: Cyber-physical
Interactor (I) An entity that exchanges data, energy, or material with the CPS but remains outside its design scope and/or control boundary.		CI: External APIs, the Internet and other networks. PI: Raw material sources; water, gas or electric mains. CPI: Humans (e.g., employees, contractors), external orgs (e.g., supply chain providers, partners, customers), technological entities (e.g., autonomous delivery and service robots).	
Trust Boundary (TB) A virtual and/or physical zone of privileged access.		CTB: Password-protected systems, encrypted files, or trusted computing environments. PTB: Physically secured areas with controlled access, locked rooms, fenced perimeters, analog safes, motor housings, machine casings. CPTB: Secured areas with both physical barriers (locks, fences) and cyber controls (authentication, surveillance monitoring).	
Store (S) Data, energy, or material at rest, distinct from its storage medium or container. Nesting is allowed.		CS: Files, databases, registry keys. PS: Raw materials, simple manufactured objects, physical keys. CPS: Smart materials, physical key cards, 3D-printed objects. <i>Note: The 3D-printed object's transformation from cyber to physical makes it vulnerable to time-dependent cyber-physical threats.</i>	
Flow (F) Data, energy, or material in motion, distinct from its enabling path, channel or medium.		CF: Function calls, network communications, data transfers. PF: Material flows, energy transfers, mechanical forces. CPF: Sensor data streams, HVAC / IoT communications, transport of smart materials or devices, cyber-physical process I/Os.	
Link (L) - New for CPFD A logical and/or physical path, channel, or medium that connects and enables Flows between CPFD elements.		CL: File formats / schema, data structures; communication ports, channels, & protocols. PL: Geographic routes, power lines, fluid pipes. CPL: RF spectrum, air (visible light / IR / acoustic transmission).	
Process (P) Activity that transforms inputs into outputs.		CP: Only digital inputs and outputs, e.g. any running code . PP: Only physical inputs and outputs, e.g. manual manufacturing, simple raw material mixing / refining. CPP: Cyber-physical inputs and/or outputs, e.g. OT processes, smart manufacturing, automated logistics, robotic assembly, adaptive environmental control, etc.	
Device (D) - New for CPFD An instantiation of computational capability and/or physical functionality for Processes and Stores; a virtually- and/or physically-embodied enabler of Processes and/or Storage in a cyber-physical system.		CD: Abstracted virtual / digital resources, e.g. virtual sensors and machines, Docker containers, digital twins, cloud compute instances, remote database servers, content delivery networks (CDN), cloud storage instances, distributed blockchain ledgers. PD: Mechanical actuators, manual valves, analog gauges, hydraulic motors, physical key storage lockboxes, material storage tanks, pressure vessels, chemical reagent containers. CPD: Embedded systems, smart thermostats, autonomous vehicles, IoT-enabled medical implants, OT actuators, desktop computers, 3D printers, smart inventory management systems, RFID-enabled storage cabinets, IoT-connected storage tanks with sensors.	

Fig. 2. CPSTRIDE Cyber-physical Flow Diagram Specification

The DFD *Process* (*P*) element represents “any running code.” This definition can be expanded to include analogous CPS processes, but we wish to distinguish between cyber-physical processes (e.g., manufacturing, refining, automated logistics) and the entities that enable them (e.g., manufacturing and refining equipment, logistic robots). We define the CPFD Process element as “activity that transforms inputs into outputs,” modify its abbreviations and keep the graphical symbol, and choose to represent Process enablers with the new CPFD Device element. The new CPFD *Link* and *Device* elements represent the paths and media that enable Flows and the entities that enable Stores and Processes, respectively. Their definitions, abbreviations, and graphical symbols can be found in Figure 2; visually, these elements combine existing graphical conventions of Trust Boundaries, Flows, and Processes.

3.2 Security Properties and Threats

Our next task is to expand the STRIDE framework’s Security Properties and corresponding Threats as needed to include CPS contexts; most of these need only minor modifications, which can be found in Figures 3 and 4. However, we find that Authentication, Confidentiality, and Information Disclosure are too narrowly data-centric for CPS, and require more substantial redefinition.

CPSTRIDE Cyber-Physical Security Properties	
Highlighted rows contain new CPSTRIDE Security Properties.	
Property	Definition
Authenticity (previously <i>Authentication</i>)	System elements (such as users, processes, devices, materials, and energy sources) are genuine and can be verified as what they claim to be. Authenticity replaces and includes the traditional <i>Authentication</i> security property for data systems while extending to the verification of physical components, materials, and energy signatures in cyber-physical contexts.
Integrity	System elements (such as data, software, firmware, hardware, materials, and energy parameters) remain unaltered and uncorrupted by unauthorized means throughout their lifecycle. This preserves the traditional data Integrity concept while expanding to include physical properties such as material composition, structural integrity, and energy calibration.
Non-Repudiation	Actions performed within the system cannot be denied by their initiator, through providing sufficient evidence of activities across cyber and physical domains. This extends beyond digital audit trails to include physical evidence trails, sensor data, surveillance records, and material verification techniques that establish accountability.
Containment (previously <i>Confidentiality</i>)	System elements (such as data, energy, and material resources) remain within their authorized boundaries and are accessible only to entities with appropriate privileges. Containment replaces and includes the traditional data <i>Confidentiality</i> security property, and more broadly represents the prevention of unauthorized cyber-physical extraction, leakage, or diversion.
Availability / Reliability	System functions, services, and resources are accessible and operational when needed, at expected performance levels. This maintains the traditional concept of digital Availability while extending to the physical reliability of components, consistent energy supply, material accessibility, and operational continuity across the cyber-physical spectrum.
Authorization	Specific entities are explicitly granted or denied permission to access, control, or modify certain system elements. This extends traditional digital access controls to include physical access rights, operational authority over equipment, material handling permissions, and energy distribution controls throughout the cyber-physical system.

Fig. 3. CPSTRIDE Security Properties Specification

The Authentication security property deals with the identification of users and software agents; we wish to expand the core meaning to include CPS analogs such as physical or cyber-physical parts or materials assumed to have come from a third-party supply chain provider. We define a new property – *Authenticity* – that includes the authentication of data entities as well as physical entities. Similarly, the Confidentiality property and corresponding Information Disclosure threat are too data-centric. The core meaning involves “(limiting) unauthorized access of valuable system elements and resources.” These can be expanded to CPS analogs, such as the extraction or diversion of physical resources by unauthorized recipients, so we define a new security property and corresponding threat – *Containment* and *Interception* – that deal with (limiting) unauthorized interception of cyber-physical entities, including data (see Figures 3 and 4).

3.3 Susceptibility Matrix

Next, we consider that the STRIDE model conventionally assumes certain types of DFD elements to be immune to specific threats; for example, DFD Data Stores and Data Flows are considered immune to Spoofing; Interactors are assumed immune to Tampering; Information Disclosure; and Denial of Service threats; and Process elements are uniquely susceptible to Elevation of Privilege (see Figure 5). In the context of secure software development these assumptions may be reasonable. However, CPSTRIDE’s inclusion of physical dimensions and novel attack vectors motivates their reexamination. In addition, we argue that although Interactors are outside of system scope and/or control, the exercise of considering the effects of attacks on Interactors can still inform system hardening. The same argument holds for all other CPFD elements and threat categories. Accordingly, we broaden the assumed susceptibility of all CPFD elements to all threat categories, and leave narrowing the matrix to future work.

4 AM Threat Modeling: CPSTRIDE vs. STRIDE

To show the advantages of CPSTRIDE over STRIDE, we perform the first two steps of the threat modeling process side-by-side. Since the third and fourth steps for both frameworks are identical, we leave them as a future exercise. The CPS to be modeled is an advanced additive manufacturing facility providing critical parts for downstream customers. The hardware setup consists of a commercial-grade bound metal deposition (BMD)-type 3D printer, a debinder, and a sintering furnace, as well as a workstation and a network router/switch.

4.1 AM System Flow Diagram Modeling

First we perform the CPSTRIDE version of Step 1, modeling the AM system with a CPFD (Figure 6). Each of the AM machines is modeled as a cyber-physical Device (CPD1-5). The router is connected via Cat5 ethernet cable to the other Devices, as well as to an enterprise network with Internet access. The networks

CPSTRIDE Cyber-Physical Threats		
Each Threat potentially violates a corresponding Security Property. In the Examples column, <i>Cyber</i> indicates threats to information, data, control signal, etc.; <i>Physical</i> indicates threats to material, energy, force, etc.; <i>Cyber-physical</i> indicates the integration or combination of the two. Highlighted rows contain new CPSTRIDE Threats.		
Threat	Definition	Examples C: Cyber P: Physical CP: Cyber-physical
Spoofing	Falsification of identity, source, or authenticity of system elements, including users, processes, signals, or physical/cyber-physical stores, undermining trust mechanisms and authentication controls within the CPS. <i>Violates Authenticity.</i>	C: Phishing, smishing, social engineering, malicious broadcast of trusted WiFi network SSID, typosquatting, deepfaking. P: Faking physical credentials, passing off counterfeit parts and materials as genuine, forging signatures on physical documents. CP: Broadcasting fake GPS to misguide autonomous vehicles or drones, injection of counterfeit OT sensor readings.
Tampering	Unauthorized modification, corruption, or alteration of legitimate cyber-physical entities including data, structures, energy flows, material compositions, or control signals, that compromises system integrity. <i>Violates Integrity.</i>	C: Modifying control logic in industrial automation software. P: Physically adjusting valve settings or equipment calibration screws. CP: Altering sensor readings through electromagnetic interference, causing the system to respond to fabricated conditions.
Repudiation	Denial of responsibility for actions within the system, either through passive rejection of accountability or active measures to destroy, corrupt, or disable auditing mechanisms or evidence trails that would establish proof of activities, legitimate or malicious. <i>Violates Non-repudiation.</i>	C: Disabling logging mechanisms to hide evidence of digital access. P: Destroying physical access records or tampering with surveillance footage. CP: Cross-domain log corruption.
Interception <i>(previously Information Disclosure)</i>	Unauthorized acquisition or monitoring of system resources, including data, energy flows, or physical materials, violating containment. Interception replaces and includes the traditional Information Disclosure threat, and <i>Violates Containment.</i>	C: Capturing sensitive control data through network sniffing. P: Physically extracting / diverting material from manufacturing processes. CP: Harvesting energy from wireless power transmission systems through unauthorized coupling.
Denial of Service	Impairment or prevention of system availability through any means that renders services, functions, or resources inaccessible or unreliable for legitimate users. <i>Violates Availability / Reliability.</i>	C: Network flooding, resource exhaustion, communication jamming. P: Blockage of moving parts; permanent damage by physical destruction, component sabotage, or irreversible physical alterations; energy disruption through power supply manipulation or battery depletion; environmental manipulation to introduce adverse conditions. CP: Creating electromagnetic interference to disrupt wireless communications and/or electronic sensors, physical obstruction of sensors/actuators.
Elevation of Privilege	Exploitation of system vulnerabilities to gain unauthorized higher-level access rights beyond assigned permissions. <i>Violates Authorization.</i>	C: Traditional privilege elevation cyber-techniques such as exploiting software vulnerabilities to gain administrative access to control systems. P: Obtaining master keys or accessing restricted physical areas without authorization. CP: Using physical access to maintenance ports to install privileged software that bypasses normal authorization controls.

Fig. 4. CPSTRIDE Threats Specification

Fig. 5. Assumed susceptibility for DFD and CPFD elements.

DFD Element	S	T	R	I	D	E
Interactor	✓		✓			
Data Flow		✓		✓	✓	
Data Store		✓	✓	✓	✓	
Process	✓	✓	✓	✓	✓	✓



CPFD Element	S	T	R	I	D	E
Interactor	✓	✓	✓	✓	✓	✓
Trust Boundary	✓	✓	✓	✓	✓	✓
Store	✓	✓	✓	✓	✓	✓
Flow	✓	✓	✓	✓	✓	✓
Process	✓	✓	✓	✓	✓	✓
Link	✓	✓	✓	✓	✓	✓
Device	✓	✓	✓	✓	✓	✓

are modeled as cyber-Interactors (CI1, CI2) and each physical connection is modeled as a cyber-physical Link (CPL1-6). Four cyber-physical Interactors are modeled: an internal designer/engineer, an operator/controller, an external material supply chain provider, and a critical downstream manufacturer (CPI1-4).

The AM process chain is broken into nine steps. Digital process chain steps from CAD design to toolpath transmission are represented by cyber-Processes (CP2-6) connected by cyber and cyber-physical Flows (CPF1, CPF2, CF1-10), while physical part production is represented by cyber-physical Processes (CPP1-4) with cyber (toolpath) and cyber-physical (material cartridge, physical part, operator input) input Flows (CF11, CPF3-12, CPF15).

Part CAD files are modeled as nested cyber-stores (CS2, NCS2.1-2.5) located on the workstation, and the AM part material cartridge is modeled as a cyber-physical store (CPS1). The physical stages of manufactured part production are modeled as four separate cyber-physical stores (CPS2-CPS5). Other modeled entities include material transport to and from the external downstream manufacturer and supply chain provider (cyber-physical Flows CPF13-14), both of which cross the primary AM Facility Cyber-Physical Trust Boundary (CPTB1); the operator/controller's quality control inspection of the AM part in the final production step (physical flow PF1); the transmission of design requirements and production parameters from the enterprise network to the human Interactors and the CAD files (cyber Flows CF12-14); and the emanation of RF and acoustic energy from the 3-D printer during operation (cyber-physical Link CPL7).

Next we perform the STRIDE version of Step 1, modeling the AM system with a DFD (Figure 7). The AM facility is modeled as a Trust Boundary (TB1), with the Internet, human designer/engineer and operator/controller, router/switch and enterprise modeled as Interactors (I1-5). The AM process chain is modeled as 9 separate Processes (P1-9) with 20 Data Flows (DF1-20). CAD files are modeled as individual Data Stores (DS1-5), and the transmission of design requirements and production parameters from the enterprise network to the humans and router are modeled by Data Flows (DF14, DF21-22), but omitted for the CAD files to avoid cluttering the diagram. Physical entities are not modeled, nor are entities which only interact with the system physically; this includes several Links, Stores, and Devices modeled in the CPFD, as well as the supply chain and downstream manufacturing Interactors.

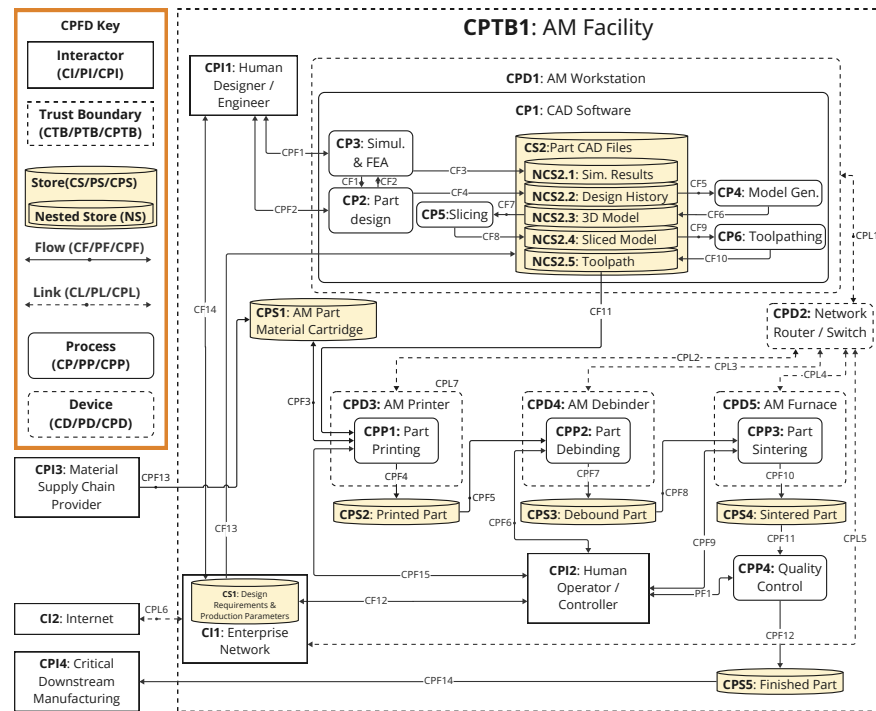


Fig. 6. Cyber-Physical Flow Diagram (CPFD) for additive manufacturing facility.

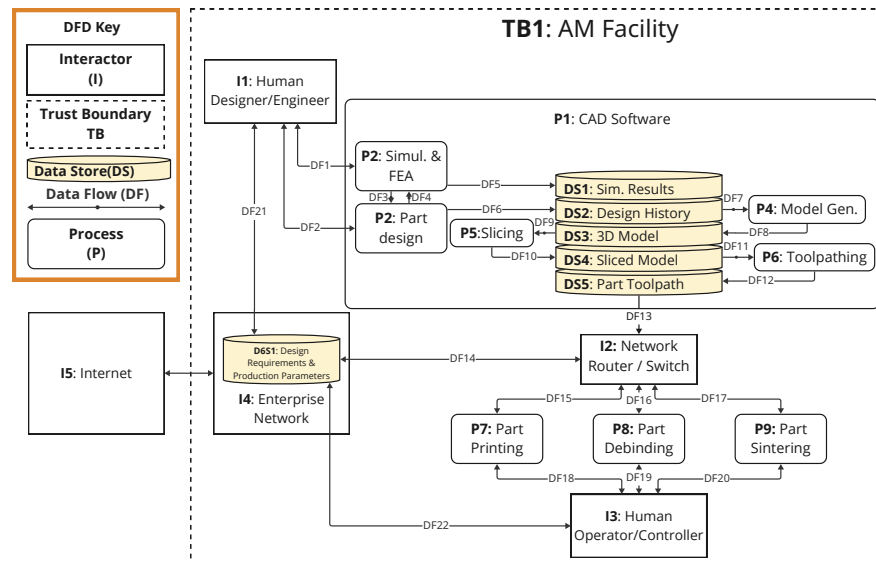


Fig. 7. Data Flow Diagram (DFD) for additive manufacturing facility.

4.2 AM System Threat Identification

Next, we perform threat identification using each framework by considering how the threat categories apply to all flow diagram elements. For instance, to identify threats against the STRIDE representation of the 3-D Model (DS3) element in the DFD, we consult the STRIDE susceptibility matrix (Figure 5) for Data Stores and find susceptibility to Tampering, Repudiation, Information Disclosure, and Denial of Service threats. We then consider how each threat might apply to the digital 3-D Model; Tampering with the 3-D Model could alter the design of the part; a Repudiation threat might involve modifying the 3-D Model’s design history; an Information Disclosure threat might represent IP theft or publicly leaking the 3-D Model; a Denial of Service threat could represent ransomware encryption or deletion of the 3-D Model. We use each applicable threat category as a brainstorming prompt to generate examples of threats to each diagram element modeled in the system. We repeat this process until all applicable threat susceptibilities have been considered for all diagram elements.

This approach generates extensive lists of threats to the system, but presents at least three challenges. First, this approach is time consuming and scales with size; larger and more complex system flow diagrams have more threats to identify. Second, this approach relies on expert knowledge; cyber-security professionals must not only be well-acquainted with the architecture and function of their system, but also consult online repositories such as MITRE ATT&CK and CVE to help identify likely cyber threats. Since there are no cyber-physical threat databases mirroring CVE for CPS, threat modeling in this domain must rely even more heavily on personal expert knowledge. A third challenge with this approach involves the ambiguity inherent in considering how a threat category might apply to an element. For instance, a particular system element might be considered the subject of a threat (i.e., the performer of a threat action), the object (target/victim) of a threat, or the instrument (used as a means to threaten another target) of a threat, or something else altogether.

We partially address the first two challenges by employing Anthropic’s Claude 3.7 Sonnet large language model (LLM) as an automated assistant threat modeler. In response to the third challenge, we adopt a posture of systematically considering each threat category / element combination from all three of the perspectives mentioned above (subject, object, and instrument).

4.3 LLM-assisted Threat Modeling

Arguably, a human expert given sufficient time would be capable of generating a threat matrix of a higher quality and more comprehensive coverage than that produced by an LLM assistant (in 2025), whether using CPSTRIDE or STRIDE. However, CPFD’s expanded consideration of physicality, devices, and paths elicits the enumeration of physical and cyber-physical elements and interactions which the DFD, by its cyber-centric nature, excludes. We expect the CPSTRIDE process, therefore, to reveal threats that would remain hidden by an equally rigorous application of the STRIDE process, whether the threat modeler is an

expert human or an LLM assistant. With this in mind, we pursue an abbreviated, first-order differential approximation of comparative STRIDE/CPSTRIDE threat modeling by instructing the LLM assistant to identify all AM system elements in the CPFD for which no corresponding DFD element exists, and then to create a CPSTRIDE threat identification matrix for the CPFD using only these elements.

Accordingly, we instruct the LLM to play the role of a subject matter expert in AM systems and threat modeling and provide it with context similar to that which a human expert might be provided, including full CPSTRIDE and STRIDE framework specifications. We instruct the LLM to identify purely cyber threats (which would also be captured by STRIDE) as well as physical and cyber-physical threats (which would not be captured by STRIDE). To resolve potential ambiguity, we instruct the LLM to systematically consider each threat category / element combination in subject, object, and instrument contexts. To limit hallucinations, we direct the LLM to identify realistic threats only and emphasize that filling each cell is not necessary. To reduce context window pressure, we instruct the LLM to work through the threat matrix carefully, completing a small number of matrix rows before awaiting human-in-the-loop approval to continue building the matrix. Finally, to reduce redundant threat identification, we give the LLM permission through instruction to group elements with similar or identical threats, and provide additional formatting instructions. All context and prompts provided to the LLM, as well as all resulting conversations, are included in our project Repository 6 for review.

The results of this LLM-assisted threat modeling, modified in format only for clarity and appearance, are found in Figures 8 and 9. These figures contain many physical and cyber-physical elements, interactions, and threats to CPS that have no standardized representation in STRIDE. For example, in the Spoofing column, highlighted cells not only identify the potential for the 3-D printer’s network identity to be spoofed (a purely cyber-threat), but also for the 3-D printer to be an instrument for physically spoofing the manufacturing source of counterfeit printed parts. Similarly, the cyber-physical AM Part Material Cartridge could be the object of counterfeiting with fraudulent ID tags / labels (physical threat), or the instrument used to introduce unauthorized physical materials into the AM system. The Material Supply Chain Provider and Critical Downstream Manufacturing recipient also receive consideration, and examples of physical subject and object spoofing threats are provided for each. Each threat identified in the matrix involves either a physical / cyber-physical element (interactor, process, store, device, flow, or path) or a threat susceptibility in the AM system which would not be captured by the STRIDE framework.

Additional examples are found in the remaining columns. The highlighted cells in the Repudiation column deal with erasure, modification, tampering, and corruption of handling logs, transfer records, chain of custody, shipping manifests, delivery confirmations, and certificates of authenticity. Although such records are often stored digitally in real-world manufacturing environments, many physical counterparts such as shipping labels and engraved or embossed

serial numbers are also widely used; both types are integral to the physical identification, handling, and tracking of manufactured physical objects. Physical and cyber-physical threats captured in the Interception column include IP theft through physical side channel emissions, physical tapping of network lines, installation of cyber-physical hardware keyloggers or packet sniffers, and physical theft and material diversion.

In the Denial of Service column, highlighted cells contain threats that include physical damage to 3-D printer via sabotage, improper maintenance, and overheating by disabling cooling systems, as well as damage to other AM system components such as ventilation systems, chemical baths or catalysts (AM debinder), gas supply lines and thermal barriers (AM furnace). In the Elevation of Privilege column, highlighted cells identify threats related to the abuse of Supply Chain Provider and Critical Manufacturing customer trust relationships to introduce unauthorized materials or components or to gain unauthorized physical access to facility or processes.

5 Discussion

The CPSTRIDE framework models physical entities, processes, interactions, and threats in CPS that the STRIDE framework cannot. The consideration of susceptibility for all CPFD elements to all types of threats allows more possibilities to be explored during the brainstorming phase of threat identification than in the conventional STRIDE approach. Another advantage of CPSTRIDE is that it allows humans to be modeled as cyber-physical Processes or even Devices (though we recommend they be modeled as Interactors); STRIDE seemingly has no satisfactory options for modeling humans. To illustrate: if humans are modeled as STRIDE Interactors (immune to T, I, D, and E attacks), then the DFD is blind to the potential impacts on CPS of human-related attacks like blackmail, bribery, social engineering, and phishing (Tampering or Information Disclosure), as well as double agency and insider threats (Denial of Service or Elevation of Privilege). Modeling humans as STRIDE Processes allows these threats to be considered, but is a clunky workaround that requires either the redefinition of humans as “any running code” or the redefinition of the STRIDE Process element - either of which would create inconsistencies in the STRIDE/DFD framework, as would redefining Interactor immunity.

STRIDE’s limitations extend to modeling non-human physical stores and flows in at least two ways. First, the modeling of physical stores and flows as data is visually and conceptually confusing, and without some visual convention for distinguishing files from parts, for example, the potential for human error in modeling and mitigating threats is increased. Second, modeling physical stores and flows as data in STRIDE bestows immunity to spoofing and elevation of privilege on both, making DFD blind to any cascading effects of counterfeiting and unauthorized access to stores and flows, in addition to bestowing immunity to repudiation (e.g., attacks on inventory systems) for physical flows (inventory in/out of controlled areas, for example).

Element	Spoofing	Tampering	Repudiation
CPD3: AM Printer	Object: Adversary could spoof printer identity in the network, causing misrouting of sensitive toolpath files. Instrument: Printer could be used to spoof manufacturing source of parts (counterfeiting).	Object: Physical tampering with printer's calibration or components to alter part dimensions or introduce defects. Object: Tampering with embedded sensors to create false readings on material deposition quality.	Subject: Printer's firmware could be modified to report incorrect logging of operations, hiding evidence of unauthorized usage. Object: Logger or operation recorder could be disabled, preventing attribution of print jobs.
CPD4: AM Debinder	Object: Spoofed control signals from unauthorized sources could initiate dangerous operations.	Object: Tampering with temperature controls to alter material properties. Object: Tampering with safety interlocks to allow opening during operation.	Subject: Manipulation of operation logs to hide unauthorized chemical or material handling.
CPD5: AM Furnace	Similar to debinder, plus: Object: Spoofing of temperature sensors to trick safety systems.	Object: Tampering with gas supply or atmosphere control systems. Object: Modification of heating profiles to affect material properties.	Subject: Alteration of sintering records to hide unauthorized operations.
CPD2: Network Router/Switch	Object: Physical spoofing of device (replacing with malicious lookalike). Object: MAC address spoofing.	Object: Physical tampering with networking hardware to create backdoors or monitoring capabilities.	Object: Tampering with physical logs or indicators of connection activity.
CPL1-CPL6: Physical Network Connections	Object: Spoofing by creating a physically similar but malicious connection pathway. Instrument: Used as a medium for spoofed transmissions.	Object: Physical tapping, splicing or modification of cables. Object: Adding unauthorized hardware inline.	Instrument: Modified physical medium could be used to alter transmission records or logs.
CPL7: RF/Acoustic Side Channel	Instrument: Could be used to inject false signals into sensors.	N/A	N/A
CP51: AM Part Material Cartridge	Object: Counterfeit material cartridges with fraudulent identification tags/labels. Instrument: Used to introduce unauthorized materials with spoofed identification.	Object: Adulterating material composition. Object: Tampering with material feed mechanisms. Object: Tampering with RFID/identification chips on cartridges.	Object: Removal or alteration of material batch tracking identifiers.
CP52-CP55: Physical Parts (Printed, Debound, Sintered, Finished)	Object: Counterfeit parts introduced into production chain. Instrument: Tampered parts could be used to falsely authenticate production source.	Object: Physical modification of parts between process steps. Object: Introduction of internal defects or weaknesses that evade detection.	Object: Removal or alteration of part identification markings/serial numbers.
PF1: Physical Inspection Flow	Object: Presenting different parts for inspection than those that were manufactured. Subject: Inspector could be impersonated by unauthorized personnel.	Object: Manipulation of physical inspection tools or gauges to incorrectly pass defective parts. Object: Tampering with part between manufacturing and inspection.	Subject: Inspector denying inspection was performed. Object: Removal or alteration of inspection stamps or markers.
CPF3-CPF12: Material/Part Movement Flows	Object: Redirecting physical materials to incorrect machines through misleading labeling or handling instructions.	Object: Modifying physical parts during transfer between process steps. Object: Swapping genuine parts with counterfeits during transfers.	Subject: Denying responsibility for part handling errors. Object: Removal of handling logs or transfer records.
CPF13: Material Flow from Supply Chain	Object: Accepting materials from fraudulent suppliers posing as legitimate vendors.	Object: Tampering with incoming materials during shipping or receiving. Object: Modifying material manifests or certifications.	Subject: Supplier denying sending defective materials. Object: Tampering with shipping records or chain of custody.
CPF14: Part Flow to Downstream Manufacturer	Object: Redirecting finished parts to unauthorized recipients through fraudulent shipping instructions.	Object: Tampering with parts during shipping or handling. Object: Swapping genuine parts with counterfeits during transit.	Subject: Denying shipping defective parts. Object: Tampering with shipping manifests or delivery confirmations.
CPF15: Operator Control Interactions	Subject: Spoofing operator identity through stolen badges or biometric workarounds. Object: Spoofed physical interfaces tricking operators into incorrect actions.	Object: Tampering with physical controls (buttons, switches) to cause incorrect operator inputs. Object: Altering physical feedback mechanisms (displays, indicators).	Subject: Operator denying initiating certain physical actions. Object: Disabling physical interaction logging systems.
CP13: Material Supply Chain Provider	Subject: Providing counterfeit materials with falsified certifications. Object: Supplier identity could be spoofed by competitors or malicious actors.	Subject: Deliberately altering material properties or specifications. Object: Supply chain systems could be tampered with to alter material specifications.	Subject: Denying knowledge of material defects or certification issues. Object: Tampering with supply chain records to hide responsibility.
CP14: Critical Downstream Manufacturing	Subject: Falsely reporting part quality issues to gain access to manufacturing facilities or processes. Object: Downstream manufacturer identity could be spoofed to intercept parts.	Subject: Requesting unauthorized modifications to physical specifications. Object: Tampering with parts after delivery but claiming they were received damaged.	Subject: Denying receipt of parts or falsely claiming defects. Object: Tampering with receiving records to hide responsibility.
CPP1: Part Printing Process	Object: Process could appear to be functioning correctly while actually using modified parameters through sensor spoofing.	Object: Physical tampering with print environment (temperature, humidity). Object: Tampering with physical layer registration or alignment.	Object: Alteration of process logs to hide evidence of abnormal printing conditions.
CPP2: Part Debinding Process	Object: Process monitoring could be spoofed to indicate proper debinding while incomplete.	Object: Tampering with physical environmental conditions affecting chemical reactions. Object: Altering solvent compositions or gas mixtures.	Object: Manipulation of process records to hide improper debinding conditions.
CPP3: Part Sintering Process	Object: Temperature or atmosphere monitoring could be spoofed to indicate proper sintering while improper.	Object: Manipulation of physical sintering environment (gases, pressure, temperature gradients). Object: Tampering with cooling rates to induce internal stresses.	Object: Alteration of thermal records to hide improper sintering conditions.
CPP4: Quality Control Process	Object: QC equipment could be manipulated to show false positive inspection results.	Object: Tampering with physical test specimens. Object: Manipulation of physical calibration standards.	Object: Falsification of physical inspection records. Object: Removal of physical defect indicators.

Fig. 8. The CPSTRIDE threat matrix identifies physical and cyber-physical spoofing, tampering, and repudiation threats in the AM system that have no standardized representation in STRIDE. Highlighted cells are special examples discussed further in the text.

Element	Interception	Denial of Service	Elevation of Privilege
CPD3: AM Printer	<i>Object: Side-channel emissions (acoustic/electromagnetic) could be intercepted to reverse-engineer part designs or manufacturing parameters. Object: Physical observation of printer operation could reveal proprietary manufacturing techniques.</i>	<i>Object: Physical damage to printer through sabotage or improper maintenance. Object: Overheating through disabling cooling systems. Object: Jamming of moving parts with foreign materials.</i>	Object: Unauthorized physical access to maintenance ports allowing privileged commands to be sent. Object: Exploitation of firmware update mechanisms to install modified firmware with elevated privileges.
CPD4: AM Debinder	Object: Unauthorized collection of process parameters through physical monitoring or tapping of control signals.	<i>Object: Physical blockage of ventilation systems. Object: Contamination of chemical baths or catalysts. Object: Damage to heating elements.</i>	Object: Exploitation of maintenance interfaces to gain control of equipment beyond operator permissions.
CPD5: AM Furnace	<i>Object: Monitoring of thermal profiles could reveal proprietary sintering parameters.</i>	<i>Object: Sabotage of gas supply lines. Object: Damage to heating elements or thermal barriers. Object: Manipulation of cooling rates to induce part stress.</i>	Object: Bypassing thermal safety systems through physical access to control circuits.
CPD2: Network Router/Switch	<i>Object: Physical tapping of network lines. Object: Installation of hardware keyloggers or packet sniffers.</i>	<i>Object: Physical damage or disconnection of network cables. Object: Signal jamming using physical proximity devices.</i>	Object: Physical access to reset mechanisms allowing reconfiguration with privileged access.
CPL1-CPL6: Physical Network Connections	<i>Object: Physical eavesdropping on network cables through electromagnetic monitoring. Object: Physical taps on communication lines.</i>	Object: Physical cutting or disconnection of cables. Object: Electromagnetic interference/jamming of communication channels.	Instrument: Unauthorized physical access to connection points could enable privileged network access.
CPL7: RF/Acoustic Side Channel	<i>Object: Passive interception of unintentional electromagnetic or acoustic emissions to extract operational data or designs.</i>	Object: Deliberate RF/acoustic jamming to disrupt environmental sensors or wireless communications.	N/A
CPS1: AM Part Material Cartridge	Object: Unauthorized sampling of proprietary material formulations.	Object: Contamination or degradation of material properties. Object: Deliberate damage to feed mechanisms.	N/A
CP52-CP55: Physical Parts (Printed, Debound, Sintered, Finished)	Object: Physical theft of parts to obtain design information through reverse engineering. Object: Unauthorized measurements or scanning of parts.	Object: Physical damage to parts. Object: Contamination with materials that could cause downstream failures.	N/A
PF1: Physical Inspection Flow	Object: Unauthorized observation of inspection techniques could reveal proprietary quality control methods.	Object: Physical interference with inspection process. Object: Damage to inspection equipment.	Object: Unauthorized physical access to inspection area could allow an attacker to approve defective parts.
CPF3-CPF12: Material/Part Movement Flows	Object: Unauthorized observation of part transfers to gather manufacturing sequence information. Object: Visual or physical access to parts during transfer.	Object: Physical blockage of material transfer paths. Object: Creating unsafe conditions that halt transfers due to safety protocols.	Object: Unauthorized physical access to handling areas could allow bypassing of material control procedures.
CPF13: Material Flow from Supply Chain	Object: Unauthorized access to proprietary material compositions during transit. Object: Interception of material specifications or order details.	Object: Blocking or delaying physical delivery of materials. Object: Damage to critical material supplies.	Object: Using compromised supply chain access to bypass material quality controls.
CPF14: Part Flow to Downstream Manufacturer	Object: Unauthorized access to parts during transit for design information. Object: Interception of accompanying documentation revealing proprietary specifications.	Object: Blocking or delaying physical delivery of finished parts. Object: Damaging parts during transit.	N/A
CPF15: Operator Control Interactions	Object: Unauthorized observation of operator actions to learn proprietary procedures. Object: Physical surveillance of authentication actions (shoulder surfing).	Object: Physical prevention of operator access to controls. Object: Creating hazardous conditions that prevent operator presence.	Object: Physical access to operator interfaces could allow execution of unauthorized commands.
CPI3: Material Supply Chain Provider	Subject: Unauthorized collection of proprietary material usage patterns. Object: Intellectual property related to material specifications could be intercepted.	Subject: Deliberately delaying or blocking critical material deliveries. Object: Disruption of supplier operations could prevent material provision.	<i>Subject: Exploiting trusted supplier relationship to introduce unauthorized materials or components into the manufacturing process.</i>
CPI4: Critical Downstream Manufacturing	Subject: Using privileged position to gather proprietary manufacturing data. Object: Customer specifications could be intercepted.	Subject: Generating excessive emergency orders disrupting production. Object: Disruption of downstream operations could prevent part acceptance.	<i>Subject: Using trusted customer relationship to gain unauthorized physical access to facility or processes.</i>
CPP1: Part Printing Process	Object: Physical monitoring of printing process for reverse engineering. Object: Acoustic or EM emissions could reveal printing patterns.	Object: Introduction of environmental contaminants disrupting layer adhesion. Object: Creating physical conditions that trigger safety shutdowns.	Object: Exploitation of physical process controls to override safety limits.
CPP2: Part Debinding Process	Object: Monitoring of process parameters to derive proprietary debinding techniques.	Object: Contamination of debinding media. Object: Disruption of ventilation or chemical containment systems.	Object: Unauthorized manipulation of safety-critical physical parameters through process access.
CPP3: Part Sintering Process	Object: Thermal imaging to capture proprietary sintering profiles. Object: Monitoring gas consumption to derive process details.	Object: Introduction of contaminants that disrupt sintering reactions. Object: Creating thermal conditions that trigger emergency shutdowns.	Object: Bypassing physical thermal interlocks to run unauthorized sintering profiles.
CPP4: Quality Control Process	Object: Observation of QC methods to learn acceptance criteria. Object: Unauthorized access to physical test results.	Object: Damaging or miscalibrating test equipment. Object: Creating conditions that prevent proper inspection.	Object: Unauthorized access to quality control areas to approve defective parts.

Fig. 9. The CPSTRIDE threat matrix identifies physical and cyber-physical interception, denial of service, and elevation of privilege threats in the AM system that have no standardized representation in STRIDE. Highlighted cells are special examples discussed further in the text.

A prominent advantage of CPSTRIDE is its ability to differentiate between cyber-physical links and flows. This, for example, allows attacks on communication protocols to be considered separately from attacks on data to be transmitted and received, as well as threats to physical communications routing (tampering, denial of service) and RF spectrum (interception). CPSTRIDE’s distinction between cyber-physical processes and devices further allows the modeling of physical threats to equipment and infrastructure posed by digital and physical sabotage.

Comparing our AM facility diagrams from the previous section, we notice that the STRIDE DFD is unable to capture many details relevant to cyber-physical operations in the AM facility environment. For instance, the cyber-physical print material cartridge flow from the external supply chain provider, and thus cyber-physical threats such as counterfeited, sabotaged, or backdoored cartridges crossing the AM facility trust boundary. Similarly, the DFD is unable to represent the downstream manufacturer, and thus is blind to the potential risks to downstream systems, human health, and financial liability associated with shipping sabotaged or defective physical parts.

Physical aspects of AM machinery are not modeled in the DFD, meaning that the threat of production delays and financial damages related to reduction of physical manufacturing capabilities are not considered or mitigated directly, nor are threats associated with IP theft via side-channel attacks on RF and acoustic energy emanation from the AM printer. Physical part production is not represented, nor are the physical cartridge or physical input flows by the operator/controller; this results in DFD blindness to the physical threats posed by digital file spoofing and tampering at different stages in the AM process chain, as well as to the physical threats of material part cartridge tampering or damage. Additionally, physical aspects of the quality control process are not considered in the DFD, and therefore neither can the associated threats be.

Threat identification during the brainstorming phase is just an early example of the potential usefulness of artificial intelligence (AI) for CPSTRIDE implementation; as these models advance in capability and accuracy, human-AI partnership in threat modeling is likely to reduce variation between different threat modelers or teams and result in reproducible threat assessment baselines. In addition, AI assistants may reduce costs and time by exploring attack chains and cascading effects, constructing and collecting from threat knowledge databases, cross-referencing similar systems and known attack patterns, and identifying threats based on emerging research or incidents.

5.1 Challenges and Limitations

Threat modeling with STRIDE is complex and time-intensive, and CPSTRIDE’s inclusion of physical entities increases this complexity and time-intensity. This, of course, is the classic modeler’s dilemma (how to balance scientific fidelity and complexity with practical usefulness and comprehensibility?) which CPSTRIDE makes no advances toward resolving.

A second limitation of the method is that although CPSTRIDE’s broadening of element susceptibility to threat categories allows additional freedom, it also can add ambiguity as to how certain entities should be modeled. Our CPFD element category definitions were designed to proactively resolve several ambiguities, but confounding exceptions undoubtedly exist and await discovery. This is acceptable for CPSTRIDE v1; future refinements of the model should seek ways to reduce ambiguity.

A third limitation is that the threats identified in our AM facility study are merely abstract conceptual representations of potential attacks that have no standard taxonomy or representation in physical threat databases to mirror their digital counterparts in CVE, ATT&CK, and Cyber Kill Chain. Creating and maintaining such resources would benefit from cooperation and partnership with the CPS security community, but would require a significant long-term effort to realize a workable solution.

A fourth barrier to adoption involves technological costs and limitations. Performing network scans with existing cyber-security tools is far quicker and more accurate than visually scanning or using other means to find physical vulnerabilities in CPS, which may cover sprawling geographic footprints and/or include inaccessible or dangerous areas. However, the continued adoption of autonomous maintenance and inspection robots may eventually make this more feasible.

There are many additional challenges related to the use of AI and LLM assistance for CPSTRIDE modeling. AI may lack deep understanding of specific industrial processes, or may miss subtle physical-world constraints and realities. Over-reliance on AI may lead to accepting threat suggestions without proper validation, over-engineering security solutions for unrealistic threats, and missing critical threats not suggested by AI. Human expertise will be required to assess and prioritize threats and mitigations into the foreseeable future.

6 Conclusion

CPSTRIDE is a comprehensive threat modeling framework for CPS extending STRIDE to address security challenges introduced by physical and cyber-physical elements. By expanding the threat modeling vocabulary to include physical and cyber-physical elements, CPSTRIDE enables security professionals to develop more comprehensive defense strategies for CPS. As Industry 4.0 technologies proliferate across manufacturing, healthcare, transportation, and energy sectors, the need for cyber-physical threat modeling becomes more urgent.

Our analysis compares CPSTRIDE and STRIDE modeling of an AM facility, demonstrating several advantages to our novel approach. The CPSTRIDE framework captures threat vectors such as side-channel attacks via acoustic and RF emanations, physical material cartridge tampering, and supply chain vulnerabilities that are invisible to traditional STRIDE modeling approaches. This illustrates the substantial security gap that CPSTRIDE helps fill.

Future research includes the application of CPSTRIDE to CPS domains beyond AM (smart grids, autonomous vehicles, medical devices, industrial control

systems) to validate generalizability and uncover domain-specific requirements. The development of automated tools for CPSTRIDE analysis could make the framework accessible to practitioners and reduce the burden of comprehensive cyber-physical threat modeling. These tools could leverage AI, LLMs, and other machine learning approaches to streamline diagram creation, threat identification, and vulnerability assessment processes.

Integrating CPSTRIDE with security frameworks and standards such as NIST Cybersecurity Framework, IEC 62443, and ISO/IEC 27001 could facilitate adoption. Its validation through case studies across different industries could help refine the framework and establish best practices for its implementation.

The continued evolution of cyber-physical systems demands equally sophisticated security approaches. CPSTRIDE represents a foundational step toward comprehensive CPS security, but its real-world impact will depend on widespread adoption, tool development, and integration with existing security practices. As CPS continue to transform critical sectors of our economy and society, frameworks like CPSTRIDE will become increasingly valuable for maintaining the security, resilience, and trustworthiness of our cyber-physical infrastructure.

Competing Interests. The authors have no competing interests to declare that are relevant to the content of this article.

Repository. The CPSTRIDE repository can be found online at the following URL: <https://github.com/DallasElleman/CPSTRIDE>

References

1. Al Faruque, M., Chhetri, S.R., Canedo, A., Wan, J.: Acoustic Side-Channel Attacks on Additive Manufacturing Systems. In: 2016 ACM/IEEE 7th International Conference on Cyber-Physical Systems (ICCPS). pp. 1–10 (Apr 2016). <https://doi.org/10.1109/ICCPS.2016.7479068>
2. Al Faruque, M., Chhetri, S.R., Faezi, S., Canedo, A.: Forensics of Thermal Side-Channel in Additive Manufacturing Systems. Technical Report 16-01, University of California, Irvine; Siemens Corporation, Center for Embedded and Cyber-Physical Systems (Jan 2016)
3. Beaman, J.J., Bourell, D.L., Seepersad, C.C., Kovar, D.: Additive Manufacturing Review: Early Past to Current Practice. *Journal of Manufacturing Science and Engineering* **142**(11), 110812 (Nov 2020). <https://doi.org/10.1115/1.4048193>
4. Belikovetsky, S., Yampolskiy, M., Toh, J., Gatlin, J., Elovici, Y.: dr0wned – Cyber-Physical Attack with Additive Manufacturing. In: 11th USENIX Workshop on Offensive Technologies. p. 16. USENIX (2017)
5. Duo, W., Zhou, M., Abusorrah, A.: A Survey of Cyber Attacks on Cyber Physical Systems: Recent Advances and Challenges. *IEEE/CAA Journal of Automatica Sinica* **9**(5), 784–800 (May 2022). <https://doi.org/10.1109/JAS.2022.105548>
6. Graves, L., King, W., Carrion, P., Shao, S., Shamsaei, N., Yampolskiy, M.: Sabotaging metal additive manufacturing: Powder delivery system manipulation and material-dependent effects. *Additive Manufacturing* **46**, 102029 (Oct 2021). <https://doi.org/10.1016/j.addma.2021.102029>

7. Gupta, N., Tiwari, A., Bukkapatnam, S.T.S., Karri, R.: Additive Manufacturing Cyber-Physical System: Supply Chain Cybersecurity and Risks. *IEEE Access* **8**, 47322–47333 (2020). <https://doi.org/10.1109/ACCESS.2020.2978815>
8. Hernan, S., Lambert, S., Ostwald, T., Shostack, A.: Uncover Security Design Flaws Using The STRIDE Approach. *Microsoft MSDN Magazine* (Nov 2006), <https://learn.microsoft.com/en-us/archive/msdn-magazine/2006/november/uncover-security-design-flaws-using-the-stride-approach>
9. Hojjati, A., Adhikari, A., Struckmann, K., Chou, E., Tho Nguyen, T.N., Madan, K., Winslett, M.S., Gunter, C.A., King, W.P.: Leave Your Phone at the Door: Side Channels that Reveal Factory Floor Secrets. In: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*. pp. 883–894. ACM, Vienna Austria (Oct 2016). <https://doi.org/10.1145/2976749.2978323>
10. Khan, R., McLaughlin, K., Lavery, D., Sezer, S.: STRIDE-based threat modeling for cyber-physical systems. In: *2017 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe)*. pp. 1–6 (Sep 2017). <https://doi.org/10.1109/ISGTEurope.2017.8260283>
11. McCormack, M., Chandrasekaran, S., Liu, G., Yu, T., DeVincent Wolf, S., Sekar, V.: Security Analysis of Networked 3D Printers. In: *2020 IEEE Security and Privacy Workshops (SPW)*. pp. 118–125 (May 2020). <https://doi.org/10.1109/SPW50608.2020.00035>
12. Rahman, H., Shafae, M.: Cyber-Physical Security Vulnerabilities Identification and Classification in Smart Manufacturing: A Defense-in-Depth Driven Framework and Taxonomy (2024), <https://arxiv.org/abs/2501.09023>
13. Saßnick, O., Rosenstatter, T., Schäfer, C., Huber, S.: STRIDE-based Methodologies for Threat Modeling of Industrial Control Systems: A Review. In: *2024 IEEE 7th International Conference on Industrial Cyber-Physical Systems (ICPS)*. pp. 1–8 (May 2024). <https://doi.org/10.1109/ICPS59941.2024.10639949>, iSSN: 2769-3899
14. Shostack, A.: adamshostack/DFD3 (Apr 2025), <https://github.com/adamshostack/DFD3>, original-date: 2017-11-05T21:20:58Z
15. Sturm, L.D., Williams, C.B., Camelio, J.A., White, J., Parker, R.: Cyber-physical vulnerabilities in additive manufacturing systems: A case study attack on the .STL file with human subjects. *Journal of Manufacturing Systems* **44**, 154–164 (Jul 2017). <https://doi.org/10.1016/j.jmsy.2017.05.007>
16. Yampolskiy, M., Graves, L., Gatlin, J., Skjellum, A., Yung, M.: What Did You Add to My Additive Manufacturing Data?: Steganographic Attacks on 3D Printing Files. In: *24th International Symposium on Research in Attacks, Intrusions and Defenses*. pp. 266–281. ACM, San Sebastian Spain (Oct 2021). <https://doi.org/10.1145/3471621.3471843>
17. Yampolskiy, M., Horvath, P., Koutsoukos, X.D., Xue, Y., Sztipanovits, J.: Systematic analysis of cyber-attacks on CPS-evaluating applicability of DFD-based approach. In: *2012 5th International Symposium on Resilient Control Systems*. pp. 55–62 (Aug 2012). <https://doi.org/10.1109/ISRCS.2012.6309293>
18. Yampolskiy, M., King, W.E., Gatlin, J., Belikovetsky, S., Brown, A., Skjellum, A., Elovici, Y.: Security of additive manufacturing: Attack taxonomy and survey. *Additive Manufacturing* **21**, 431–457 (May 2018). <https://doi.org/10.1016/j.addma.2018.03.015>